

VASANTHKUMAR N

Mechanical Engineering | B.E. Final Year | Kamaraj College of Engineering & Technology

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Tamil Nadu, Madurai-625022

Portfolio: <https://vasanth-nm.github.io/portfolio>

OBJECTIVE

Final-year B.E. Mechanical Engineering student with a strong foundation in CAD modelling, FEA simulation, and IoT-based systems. Seeking an entry-level Design Engineering or Manufacturing Engineer role to apply skills in Fusion, ANSYS, and hands-on fabrication for innovative product development.

EDUCATION

Institution	Course	Percentage	Year
Kamaraj College of Engineering & Technology	B.E. Mechanical Engineering	CGPA: 7.75 (Up to semester-5)	2026
Amudham Matric Hr. Sec. School	HSC	71%	2023

TECHNICAL SKILLS

CAD Software	AutoCAD (2D), Creo, Fusion, CATIA V5
Simulation / FEA	ANSYS Workbench (Static Structural, Thermal, Modal)
3D / Creative	Blender, DaVinci Resolve
Manufacturing	CNC Programming, GD&T, Metrology

PROJECTS

Digital Twin-Based Predictive Maintenance System — ongoing Final Year Project (2026)

- Developing a Digital Twin framework integrating ANSYS FEA, ESP8266 IoT, Firebase, Unity, and Machine Learning for real-time health monitoring of rotating machinery.
- Building a real-time data acquisition system using ADXL345 vibration and DS18B20 temperature sensors mounted on motor bearing housings.
- Implementing a Random Forest machine learning model to detect and classify bearing faults, imbalance, and shaft misalignment for predictive maintenance.
- Designing a Unity-based 3D digital twin dashboard synchronized with live sensor data to provide an affordable predictive maintenance solution for MSMEs.

Design & Fabrication of a Compact Heat Recovery System — 3rd Year Project (2024–25)

- Designed a shell-and-helical-coil heat exchanger for waste heat recovery Fusion.
- Performed thermal and structural analysis in ANSYS Workbench to validate design parameters.
- Fabricated and tested the prototype; measured heat transfer effectiveness and pressure drop across operating conditions.

- Optimised coil geometry and shell dimensions to achieve target effectiveness under compact form-factor constraints.

IoT / RFID-Based Real-Time Sewing Performance Tracking System — Industrial Project, Pentagon Apparels Unit II, Virudhunagar

- Implemented an RFID-based system to track operator-level sewing output in real time on the production floor.
- Integrated ESP8266 Wi-Fi modules with RFID readers and a live dashboard for real-time performance monitoring.
- Reduced manual data-entry errors and enabled supervisors to identify bottlenecks quickly.

Dynamic Speed Bumps — Thinkathon, KCET, Virudhunagar

- Presented a concept for adaptive speed bumps that flatten for emergency vehicles while remaining raised for regular traffic.

ACHIEVEMENTS

- 2nd Prize — Technical Quiz, Yuvasakthi 2K26 National Level Technical Symposium, NPR College of Engineering & Technology, Natham (Mar 2026).
- 3rd Prize — SkyForge event, Euphoria Techno Management Meet, Kalasalingam Academy of Research & Education (Mar 2026).
- NPTEL Elite Certificate — Product Design & Manufacturing, IIT Kanpur; (Jan–Apr 2026).

INTERNSHIPS

Swing Setter — Paint Shop Department Internship (15 Days)

- Observed end-to-end paint shop operations including surface preparation, coating application, and curing stages.
- Studied quality control checkpoints for finish consistency and defect identification.
- Gained practical exposure to industrial safety protocols and workflow sequencing in a production environment.

Pentagon Apparels Unit II, Virudhunagar — IoT/RFID-Based Real-Time Sewing Performance Tracking System

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CERTIFICATIONS & TRAINING

NPTEL Elite — Product Design & Manufacturing	IIT Kanpur / SWAYAM, Jan–Apr 2026
ISTE Student Member (SM No. 11709)	Indian Society for Technical Education, 2025–2027
Industrial Internet of Things (IIoT) Hands-on Training	SASTRA University, Thanjavur
CATIA V5 Certification	Kamaraj College of Engineering & Technology
Advanced Creo (3D CAD)	Udemy
CNC Programming	NTTF, Bangalore

SOFT SKILLS & INTERESTS

Problem-solving • Creative Thinking • Team Collaboration • 3D Part Modelling • CAD/CAM Automation